

Chasing the Rush: How horror games trigger adrenaline and fuel Fear-Inducing elements

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ABSTRACT

Fear has been capitalized through horror video games, driving players to develop a craving for adrenaline-fueled fear, fostering fear-inducing elements, and leading to unhealthy behavioral patterns. This study examined how fear and adrenaline are triggered in horror video games, influencing play intention and actual gameplay, and how these experiences contribute to fear-inducing elements analyzed based on their stimulus-organism-response (S-O-R) aspects. The objective was to determine significant predictors of actual gaming of horror video games by analyzing data from 1,015 valid players using the Partial Least Squares Structural Equation Modeling (PLS-SEM) approach with SOR-based analysis. Findings revealed that fear-inducing elements strongly influenced psychological response and emotional value. In contrast, game design elements primarily impacted perceived control and play intention. It was revealed that both psychological and physiological responses contribute to emotional value, while cognitive processing played a more dominant role than physiological arousal in shaping emotional impact. In accordance, play intention strongly predicts actual gameplay, but excessive fear-inducing elements can reduce actual gameplay. These results highlight the need for game designers and developers to balance fear elements with adaptive difficulty, strategic pacing, and perceived control to enhance immersion while being physiologically, and psychologically safe. Horror game players can use these insights to manage their expectations and experiences effectively. For gaming industry marketers and publishers, responsible promotion of horror games should emphasize thrilling engagement and psychological safety, ensuring sustainable and innovative gaming practices.

1. Introduction

Horror video games are now at the forefront of this rise, offering adrenaline-pumping experiences that immerse players in fear-driven narratives and spine-chilling challenges [1]. C.T. Lim [2,3] mentioned a significant growth in video game sales during March 2020, with digital game sales increasing by 57 %, physical games by 82 %, and game consoles by an impressive 155 %. Bashir [4] and Marciano [5] revealed that the Philippines led in video game usage, with the highest population of gamers, boasting a gaming penetration rate of 95.8 % in 2022. As the momentum of the video game industry continues to build, global online game user penetration is projected to reach 44.6 % or approximately 53.4 million users by 2027. Moreover, the industry's revenues are expected to climb to around \$521 billion by the same year [6].

Building on the industry's exponential growth, Graja et al. [7] indicated that horror video games have carved out a unique and

increasingly popular niche, captivating players with their ability to blend fear, suspense, and storytelling into unforgettable experiences. These games utilize dark and atmospheric settings, jump scares, and psychological tension to create an intense sense of dread that engages the sympathetic nervous system, triggering heightened heart rates and adrenaline surges [7,8]. The genre's appeal lies not only in its ability to provoke visceral reactions but also in the creative freedom it affords developers to subvert expectations, challenge conventions, and craft emotionally rich narratives that amplify the impact of each scare [9]. Horror games span diverse styles, from fast-paced thrillers to slow-burning psychological terrors, all of which are designed to test the player's courage and resilience. By immersing players in macabre masterpieces, the genre offers an addictive combination of fear and excitement, solidifying its status as a key driver of the video game industry's continuing evolution [10].

Fear, as a fundamental human emotion, has long been a powerful

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tool for engaging audiences [11], and horror video games have masterfully capitalized on its visceral and universal appeal [12,13]. Rooted in the psychology of fear, these games tap into the primal fight-or-flight response, creating immersive experiences that players find both thrilling and addictive [14,15]. The commercial success of the genre is partly attributed to the phenomenon of “fear addiction,” where the adrenaline rush and heightened sense of alertness triggered by fear become sources of pleasure [16,17].

Studies suggest that this paradoxical enjoyment stems from the controlled and safe environment of video games, allowing players to confront their fears without real-world consequences [18,19]. Developers further exploit this by crafting narratives and mechanics that exploit deep-seated fears, whether through themes of isolation, the unknown, or mortality, ensuring players remain captivated. This marketization of fear not only drives the genre’s popularity but also demonstrates its ability to resonate across diverse demographics, reinforcing the horror game’s role as a dynamic and profitable segment of the video game industry.

While the adrenaline rush triggered by fear can provide an exhilarating and thrilling experience, excessive exposure to such heightened states can have detrimental effects. Prolonged activation of the fight-or-flight response may lead to physical and psychological exhaustion, as the body is consistently pushed into a state of high alert [20]. Chronic engagement with intense fear, particularly in horror games, has been linked to increased stress, anxiety, and even burnout [21,22]. This craving for adrenaline-fueled fear can drive players to seek progressively extreme and disturbing experiences, potentially desensitizing them to the genre’s intensity and fostering unhealthy behavioral patterns.

Over time, what begins as a controlled form of entertainment may evolve into a compulsive need for constant stimulation, blurring the line between enjoyment and dependence [16,23]. In some cases, as per Behrmdtz et al. [24], players may struggle to break free from the cycle of fear appraisal, relying on adrenaline surges for emotional regulation. The long-term consequences of such reliance can include heightened anxiety, irritability, and difficulty managing stress in daily life.

Thus, there is a clear gap in exploring how fear and adrenaline are triggered in horror video games and how these experiences contribute to fear appraisals from fear-inducing elements. The importance of understanding the effects of adrenaline rushes in gaming can provide valuable insights into player behavior, contributing to the development of more engaging and immersive gaming experiences that responsibly manage the balance between fear and enjoyment. The overall response from emotional, psychological, and behavioral aspect of players have not yet been thoroughly and simultaneously assessed among related studies but has been individually assessed to either affect or is significant. To which, the need for establishment to understand how controlled, entertainment, enjoyment, and/or dependence response among players on horror games should be conducted.

This study applied the Stimulus-Organism-Response (SOR) as the study’s framework to comprehensively understand the dynamics of fear and adrenaline in horror gaming. The SOR model, which examines the relationship between external stimuli, internal cognitive and emotional processes, and resulting behavioral responses, is particularly suited for investigating the psychological effects of horror game design elements on players. The originality of this study lies as being the first to examine how horror games trigger adrenaline and fuel fear-inducing elements using partial-least-squares structural equation modeling (PLS-SEM) with SOR-based analysis.

Specifically, this study aimed to achieve the following objectives: (1) To investigate how fear-inducing elements in horror games influence physiological responses, psychological responses, emotional value, and perceived control in players; (2) to explore the impact of game design elements on players’ psychological responses, emotional value, and perceived control; (3) to assess the relationship between physiological responses, psychological responses, emotional value, and perceived control with players’ play intentions; (4) to determine how players’ play

intentions affect their actual gameplay behavior; and (5) to determine how horror games trigger adrenaline and fuel fear appraisals from fear-inducing elements and create implications, both theoretical and practical, for actual applications. This study aimed to offer a deeper understanding of how specific game elements and player reactions drive engagement and influence gameplay behavior in the context of horror gaming.

The significance of this study lies in its potential to enhance the understanding of how horror games trigger adrenaline and fuel fear-inducing elements, contributing to the broader field of gaming psychology, and the video game industry. Understanding the effects of adrenaline rushes and fear-inducing elements in gaming can provide valuable insights into player behavior, contributing to the development of more engaging and immersive gaming experiences that responsibly manage the balance between fear and enjoyment. This study is highly relevant to the United Nations Sustainable Development Goals (SDG), specifically SDG 3 which focuses on ensuring healthy lives and promoting well-being. As explained by Takemoto et al. [25], game designs and developers are expected to develop a platform to change consumer behavior, enhance overall wellbeing for all, and establish decision-making and respect to problems. By exploring how fear and adrenaline affect players, the research can offer valuable information on how to design games that are both thrilling and psychologically safe, reducing the potential negative effects of excessive fear or stress.

Additionally, by understanding the role of fear and adrenaline in player engagement, this study supports SDG 4, which promotes inclusive and equitable quality education and lifelong learning. The access to knowledge, strategy build-up, and skill development may enhance consumer’s overall learning [26]. The findings can inform game developers on how to create games that are not only entertaining but also contribute to cognitive and emotional development. Furthermore, this study aligns with SDG 9 which encourages innovation and resilient infrastructure by exploring the potential of psychological and physiological mechanisms in game design. It advocates for innovation in the gaming industry, where understanding player behavior and responses can lead to the development of more immersive, impactful, and responsible gaming technologies. As indicated by Myers and Murabit [27], the innovative actions in technology would help establish an industry that is sustainable, while also bridging the gap among the digital divide among consumers [26]. Establishing these areas of SDG could create a more sustainable transformation that helps the development of consumers.

2. Related studies and Conceptual framework

2.1. Related studies

Existing studies have significantly contributed to understanding fear and engagement in horror gaming, notable gaps remain. For example, Moschovitis and Denisova [28] explored the impact of biofeedback-controlled dynamic difficulty adjustment (DDA) in horror games, showing that it enhanced intrinsic motivation and perceived competence. However, their study primarily focused on enjoyment and challenge rather than fear’s specific psychological effects, leaving unexplored how physiological data integration impacts fear appraisals or long-term behavioral patterns. Yan [17] emphasized the interplay between fear and immersion in horror survival games but relied heavily on user comments and surveys, limiting insights into physiological or behavioral impacts. Furthermore, the focus on two popular games may restrict the generalizability of findings to other games in the genre, particularly those with less intense or unconventional fear-inducing mechanics. Andersen et al. [1] examined fear and enjoyment in haunted attractions and identified an inverted-U relationship between arousal and enjoyment. While their findings are relevant to understanding recreational fear dynamics, the study’s scope is limited to physical attractions and does not account for how similar principles

translate to virtual environments like horror games.

Smith [21] highlighted tension-creating mechanics like resource management and storytelling but did not explore their effects on psychological resilience or sustainable fear experiences. Lastly, Tomé et al. [22] investigated psychological horror games' potential mental health benefits but, with a small sample size, overlooked how gameplay mechanics contribute to these outcomes. The study also failed to address the broader implications of sustained tension and anxiety on players' psychological well-being. Collectively, these studies underscore the complexity of fear in gaming but reveal critical gaps in understanding its long-term effects, implications for player behavior, and the balance between enjoyment and fear appraisals. Addressing these gaps is crucial for designing responsible, immersive gaming experiences that align with psychological well-being.

2.2. Conceptual framework

The Stimulus-Organism-Response (SOR) framework provides a structured approach to examining the interplay between external stimuli, internal processes, and player behavior in the context of this study, on how horror games trigger adrenaline and fuel fear-inducing elements. This study contributes to the field by introducing a focused exploration of the fear-inducing phenomenon in horror games, a perspective that has not yet been fully addressed in existing applications of the SOR framework.

Originally developed to explain consumer behavior [29], the SOR framework has been widely adopted across various fields, including gaming, for its ability to capture dynamic interactions between stimuli, internal responses, and actions [30,31]. Kalaiarasan et al. [32], for example, applied it to investigate behavioral intention toward Extended Reality (XR) gaming, highlighting the influence of perceived ease of use, enjoyment, usefulness, and value on behavioral intentions, solidifying this study's SOR framework's relevance for exploring factors influencing player engagement and responses.

Fig. 1 was designed to examine the relationships between 8 key constructs through 12 hypotheses, offering a comprehensive lens to explore the interplay between fear-inducing stimuli, player responses, and gameplay behavior. The Stimulus (S) component comprises 2 constructs: Fear-Inducing Elements and Game Design Elements, representing the external factors that evoke player engagement. The Organism (O) component represents internal processes and includes 4 constructs: Physiological Response, Psychological Response, Emotional Value, and Perceived Control. The Response (R) component represents player actions and includes 2 constructs: Play Intention and Actual Gameplay.

Perceived Control. Notably, Physiological Response, Psychological Response, and Emotional Value are grouped together to highlight their interconnected nature. Lastly, the Response (R) component captures player actions and is represented by 2 constructs: Play Intention and Actual Gameplay.

Fear can be caused by a variety of factors, including genetic predispositions, cultural experiences, environmental factors, physiological reactions, and psychological state [33]. In the context of games, fear-inducing elements are often integrated into gameplay to evoke heightened emotional and physical responses from players. Fear-inducing elements in horror games, such as jump scares, eerie sound effects, and unsettling visuals, are designed to elicit intense emotional and physiological responses. These stimuli trigger a fight-or-flight reaction, causing measurable changes in heart rate, skin conductivity, and breathing patterns [33–35].

According to Németh and Szűcs [36], playing a gory fighting game like *Mortal Kombat* increases blood pressure and decreases heart rate compared to a logic-based game like *Tetris*. Participants also reported more positive emotions on the gory aspects of *Mortal Kombat*, indicating that fear-inducing elements in horror (like the VR horror games from Zhang et al. [33] or the jump scare video games by Villarreal [35] or violent games (as observed from the study of Németh & Szűcs [36]) can cause notable physiological changes compared to logic-based games. Further supporting this notion, a study by Kovács [37] explored the “horror paradox,” where fear, traditionally seen as negative, can be enjoyable to others due to the dopamine release following short bursts of fear. This controlled fear response can provoke strong physiological reactions while simultaneously creating pleasurable emotional experiences [38]. Andersen et al. [1] further examined the physiological responses of participants in a simulated fear environment using heart rate monitors, revealing an inverted U-shaped relationship between fear and enjoyment, with moderate fear yielding optimal responses. This highlights how fear-inducing elements in games significantly affect both physiological and psychological experiences.

Waili [16] noted that fear-inducing elements evoke psychological responses like anxiety, distress, and heightened alertness, with players perceiving these stimuli as threats, often leading to emotional discomfort or excitement. Yi and Chung [39] and Perez-Reyes [40] further highlighted that the psychological impact of horror games is more pronounced in individuals highly sensitive to fear stimuli, with responses such as tension or nervousness varying based on personality traits. Therefore, the following hypotheses are defined:

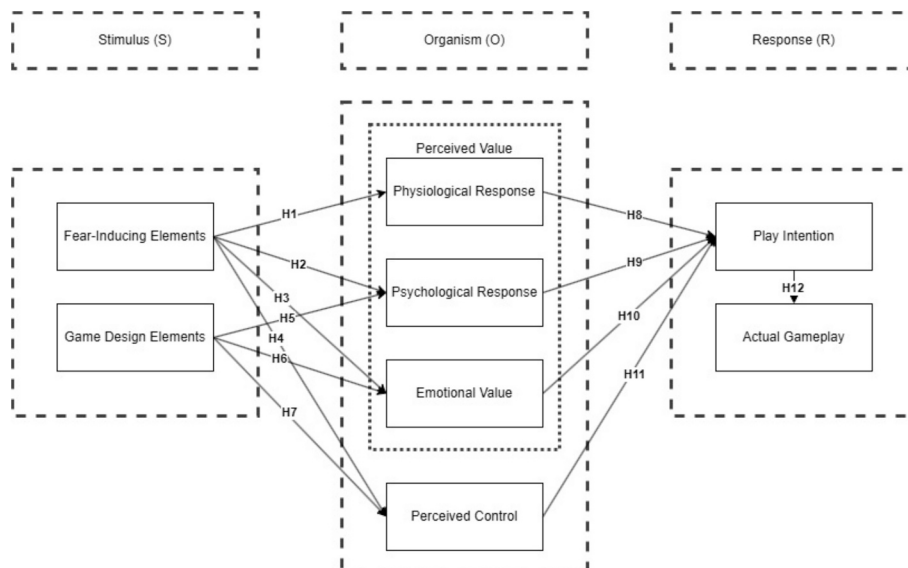


Fig. 1. Conceptual Framework.

H1. Fear-inducing elements have a significant effect on physiological response.

H2. Fear-inducing elements have a significant effect on psychological response.

Fear enhances the emotional value of video games by immersing players in experiences shaped by elements like dark settings, jump scares, and eerie sound effects [23]. These elements evoke intense reactions, such as fear and suspense, contributing to players' emotional connection, including feelings of excitement, satisfaction, and tension [33–35,41]. Tomé et al. [22] found that psychological horror games, like *Silent Hill 2*, deepen emotional value through empathy and engagement with characters. Players formed cathartic connections, experiencing emotional value beyond fear as they related to characters' struggles, emphasizing how narrative depth enhances gaming experiences.

In recent years, the impact of fear-inducing elements in horror games on players' perceived control has become a topic of growing interest within the field of game research [19]. The perception of control is a critical factor in horror games, influencing how players navigate through fear-inducing scenarios. Ferreira et al. [42] found that players' perception of control in horror games, such as making decisions or escaping threats, empowers them to manage fear. Similarly, Lemmens et al. [43] further proved that players' perceived control over the game environment can mitigate the intensity of fear and anxiety. However, fear-inducing elements in the game, such as unpredictable events or encounters with terrifying entities, can challenge this sense of control, leading to heightened tension and fear [22,41]. These findings suggest that fear-inducing elements can both disrupt and reinforce players' perceived control, influencing their overall experience and emotional engagement. Therefore, the following hypotheses are defined:

H3. Fear-inducing elements have a significant effect on emotional value.

H4. Fear-inducing elements have a significant effect on perceived control.

Game design elements play a crucial role in shaping the psychological response of players to a horror game [33]. A study by Smith [21] explored the crucial role of game mechanics, atmospheric design, sounds, and ambiance in eliciting strong emotions, particularly tension and stress. Similarly, the study of Qi and Jeanhun [44] emphasized how character design in side-scrolling horror puzzle games affects player emotions. By developing a prototype with varying gameplay scenarios, their study highlighted how the effective integration of these elements can enhance the player's emotional experience, and psychological response. Elements like surprises, dark atmospheres, and unpredictable gameplay heighten fear and anxiety by disrupting players' sense of control [13,34,35,42].

In the horror game genre, the use of specific game design elements has been shown to have a significant effect on the psychological responses of players. Elements such as lighting, sound, and environmental design can influence the way players respond and agency within the gaming world, creating a more immersive and engaging experience. Players in dimly lit, eerie environments reported lower perceived control compared to those in brighter, more open settings [34,35,42,45]. Therefore, the following hypotheses are defined:

H5. Game design elements have a significant effect on psychological response.

H6. Game design elements have a significant effect on emotional value.

Fouché's [46] study on the minimalist game design of *Iron Lung* found that elements such as obfuscation, clumsy mechanics, and slow pacing can intensify players' unease and immersion. Derived from player reviews and video analysis, the study suggested that simpler gameplay mechanics can effectively create tension and emotional responses, with a player's sense of control being key to maintaining engagement in horror games [43]. Similarly, a study by Cao et al. [45] found that players who played games with high-quality sound effects

and music reported lower levels of perceived control compared to players who played games with simpler sound design. Furthermore, Griffith and Sharpe [14] proved that the relationship between exposure to specific game design elements, such as lighting, sound, and environmental design, and the perceived control of players in horror games is consistently found to be significant, leading to increased immersion and satisfaction. Therefore, the hypothesis is defined:

H7. Game design elements had a significant effect on perceived control.

The physiological and psychological response of players to gameplay in horror games can significantly impact their play intention and motivation to continue playing [21,22,35]. Zhang et al. [47] found that fear-inducing stimuli in VR horror games trigger strong physiological responses, influencing player behavior and coping strategies. These responses, like increased heart rate or sweating, can enhance immersion for some while deterring others. The intensity of these physiological reactions directly impacts play intention, either reinforcing or hindering engagement. Similarly, Navarro et al. [48] explored how biofeedback, such as heart rate or electrodermal activity, can adapt gameplay and enhance engagement. However, poorly managed responses, leading to stress or frustration, can reduce players' intent to continue playing.

In addition, Yan [17] found that psychological responses like fear, anxiety, excitement, and immersion significantly influence players' play intention in horror games. Strong emotional reactions, such as excitement or fear, motivate players to continue for the escapism and adrenaline, while overcoming challenges can enhance satisfaction and play intention. Hawco [49] highlighted how elements like haptic mechanics, play conditioning, and coding heighten fear and tension, creating an immersive experience. For some, this cycle of challenge and fear is addictive, while for others, it can deter further engagement, reflecting the complex relationship between psychological response and play intention. Zhang [50] also found that players' psychological responses, like fear, anxiety, and satisfaction, shape their gaming experience, with thrill-seekers finding intense emotions rewarding, while others may withdraw. Luong and Holz [51] explored how transient psychological states, such as fear and frustration, trigger distinct physiological responses that influence player engagement. Based on these findings, the following hypotheses are defined:

H8. Physiological response has a significant effect on play intention.

H9. Psychological response has a significant effect on play intention.

Emotional value in horror games goes beyond fear, encompassing emotional engagement through storytelling, character empathy, and immersive design [22,33,35]. Vivekanandhan et al. [52] found that horror games elicit heightened cognitive engagement compared to calm games, fueling player excitement and engagement. This supports the idea that horror games, with their stimulating content, sustain attention and increase play intention.

In addition, Verran's [53] study highlighted the role of self-control and roleplaying in games, showing that identification with avatars, like in *Celeste*, enhances engagement through mastery and self-exploration. This duality of control motivates players to invest more time, boosting immersion and play intention. While Moschovitis and Denisova's [28] study on biofeedback-controlled dynamic difficulty adjustment (DDA) in the *Caroline* game revealed that adaptive mechanics based on biometric data enhance perceived control, increasing intrinsic motivation by balancing challenge and mastery. Therefore, the following hypotheses are defined:

H10. Emotional value has a significant effect on play intention.

H11. Perceived control has a significant effect on play intention.

On the other hand, Flanagan's [54] study proved that play intention, or the desire and motivation to play a game, significantly impacts actual gameplay. Players with strong play intention are more likely to invest time and effort, engage in challenging and immersive gameplay, and persist through difficulties. Conversely, players with low play intention may engage less, avoid challenges, or give up quickly when faced with obstacles. Christopher and Leuszler [18] found that players with higher

play intention in a horror game were more likely to interact with game objects and characters and explore the game world extensively. Similarly, Marak [55] observed that players with higher play intention exhibited greater problem-solving and strategic thinking and experienced heightened immersion and presence in the game world. These findings suggest that play intention not only motivates gameplay engagement but also shapes players' actions and cognitive approaches, such as problem-solving and immersion. Therefore, it was hypothesized that:

H12. Play intention had a significant effect on actual gameplay.

3. Methodology

3.1. Participants

The study included participants aged 18 and older, residing in the Philippines – a country with a 95.8 % gaming penetration rate as of 2022 [4,5]. The respondents, all active horror game players, were selected to meet the study's criteria using purposive sampling technique, ensuring a focus on those with relevant gaming experience. This purposive sampling approach aimed to provide a nuanced understanding of the demographics and behaviors associated with horror game engagement [56].

An online dissemination of the survey questionnaire among social media groups in Facebook, Messenger, Viber, and Instagram was utilized alongside face-to-face dissemination of survey questionnaire using a QR code. Participants voluntarily consented to contribute, with assurances that their responses would be used solely for academic purposes, kept safe, and would not be disseminated (FM-RC-22-02-01). This method ensured a diverse representation of the gaming population.

Table 1 presents a comprehensive overview of the participants' demographics and their engagement with horror games. With 1015 respondents, the survey provides a well-rounded sample that offers key insights into the demographic characteristics, gaming behaviors, and

preferences of horror game players.

The participants of this study included a diverse group in terms of gender, with a majority identifying as female (64.24 %) while 35.76 % were male. This gender distribution highlights the increasing presence of female gamers in the gaming community. Over recent years, the gaming industry has seen a rise in female players, with women now comprising a significant portion of the gaming audience [57,58]. This shift reflects broader social changes, where gaming is no longer perceived as a male-dominated activity but rather as an inclusive form of entertainment enjoyed by all genders [59]. The higher female representation in this study suggests that women are not only engaging with gaming but are also becoming a powerful demographic influencing gaming trends and preferences.

Age-wise, the study had a substantial concentration of participants in the 18–24 years old (44.53 %) and 25–34 years old (42.17 %) categories, indicating that young consumers and young adults are the primary audience for horror games. These age groups may be more inclined towards the excitement and challenge that horror games offer, as well as the engagement with gaming platforms that are more accessible and popular among younger populations. A smaller proportion of respondents were from older age brackets, with only 8.67 % falling in the 35–44 years old group and an even smaller representation from 45 years old and above (4.63 %). This demographic distribution shows that younger individuals are more likely to participate in digital entertainment like gaming due to their greater familiarity and comfort with technology and entertainment options [60].

Regarding residential areas, the majority of participants resided in urban areas (65.71 %), which is consistent with the global trend of greater access to technology and gaming resources in cities [61]. The respondents were mainly undergraduates (64.24 %), with a smaller portion holding senior high school diplomas (11.13 %), graduate degrees (19.90 %), and postgraduate degrees (4.73 %). This suggests that a majority of the participants were in their early academic or career stages, possibly balancing gaming with education or part-time work. The relatively high proportion of undergraduates may reflect the ease with which younger adults engage in gaming, as well as the significant role of gaming as a form of entertainment during this life stage [62].

The frequency and duration of playing horror games revealed a high level of engagement among participants. A large proportion of the sample played horror games either daily (25.42 %) or weekly (41.28 %), suggesting that horror gaming is a consistent form of entertainment for many participants. Additionally, 62.66 % of participants reported spending between 1 and 5 h per session playing, demonstrating that horror gaming is not just a brief pastime but an activity that often extends over substantial periods, with a smaller portion (5.62 %) playing more than 10 h at a time.

In terms of preferred gaming platforms, mobile phones were the dominant choice for 60.00 % of participants, followed by PC (21.28 %) and consoles (12.32 %). This preference can be attributed to the affordability and accessibility of mobile devices, making them an attractive option for many gamers [63]. Mobile phones offer a convenient and cost-effective way for individuals to enjoy gaming, particularly in countries with diverse economic backgrounds [64,65]. This trend aligns with the increasing popularity of mobile gaming, particularly among younger audiences who prioritize convenience and affordability in their entertainment choices [66].

Lastly, when it comes to adrenaline-seeking tendencies in horror games, 43.45 % of participants reported being "very much" inclined towards seeking out thrilling experiences, while 36.65 % were somewhat inclined. This indicates a strong desire for the excitement and fear-inducing elements that are characteristic of horror games [10,67].

3.2. Questionnaire

To collect data for this study, a self-administered questionnaire was crafted based on an extensive examination of existing literatures and

Table 1
Participants' Descriptive Statistics (n = 1 015).

Characteristics	Category	N	Percentage
Age	18–24 years old	452	44.53 %
	25–34 years old	428	42.17 %
	35–44 years old	88	8.67 %
	45–54 years old	30	2.96 %
	55 years old and above	17	1.67 %
Sex	Male	363	35.76 %
	Female	652	64.24 %
Residence	Urban	667	65.71 %
	Rural	348	34.29 %
Highest Educational Level Achieved	Senior High School	113	11.13 %
	Undergraduate	652	64.24 %
	Graduate	202	19.90 %
	Postgraduate	48	4.73 %
Frequency of Playing Horror Games	Daily	258	25.42 %
	Weekly	419	41.28 %
	Monthly	279	27.49 %
	Rarely	59	5.81 %
Session Length in Playing Horror Games	Less than 1 h	137	13.50 %
	1–5 h	636	62.66 %
	6–10 h	185	18.23 %
	More than 10 h	57	5.62 %
Preferred Platform for Playing Horror Games	PC	216	21.28 %
	Console (e.g., PlayStation, Xbox)	125	12.32 %
	Mobile	609	60.00 %
	Handheld (e.g., Nintendo Switch)	65	6.40 %
Adrenaline-Seeking Tendencies in Horror Games	Very Much	441	43.45 %
	Somewhat	372	36.65 %
	Neutral	161	15.86 %
	Not Much	33	3.25 %
	Not at All	8	0.79 %

distributed through Facebook Groups, Viber, and Messenger between December 2024 and February 2025. The questionnaire was structured into two sections: the first gathered demographic details of the participants, while the second explored critical variables grounded in the Stimulus-Organism-Response (SOR) framework as shown in Fig. 1. The questionnaire consisted of 28 items, each measured on a 5-point Likert scale ranging from “1” (Strongly Disagree) to “5” (Strongly Agree) [68]. The demographic data collected is summarized in Table 1, while detailed question items are outlined in the supplementary materials. This structure aimed to provide a comprehensive understanding of how fear-inducing elements, internal processes, and gameplay responses interact, offering valuable insights into the dynamics of horror gaming engagement, approved by the university ethics committee board (FM-RC-22-01-01) prior to dissemination.

Prior to full SEM assessment, the collected sample was tested for normality (Table 2) using the Shapiro-Wilk test. It could be seen that the results were within the ± 1.96 threshold, implicating that output is within the acceptable range [69]. Moreover, the common method bias using the Harman’s Single Factor test was utilized – resulting to 6.796 % total variance and only 24.27 % cumulative and percent of variance. Following the suggestion of Podsakoff et al. [70], this should be well below the 50 % threshold to indicate no method bias presence.

3.3. Structural Equations Modelling

This study employed the PLS-SEM using the SmartPLS 4.0 software to analyze complex relationships among latent variables. PLS-SEM has been effectively applied in similar gaming research, such as Chen and Liu [71], who analyzed students’ acceptance of architecture-related video games using the simplified UTAUT2 model. This approach is relevant to our study as it highlights the ability of PLS-SEM to examine complex relationships between player behavior, motivation, and engagement, which can be applied to understand the psychological effects of fear and adrenaline in horror gaming. In another example, Liew et al. [72] utilized PLS-SEM to investigate how hedonic and utilitarian expectations influenced continuance intentions in online gaming, further reinforcing the utility of SEM for behavioral research.

The strength of SEM lies in its capacity to handle various data types and scales—nominal, ordinal, and interval—provided there is a robust

framework for testing intricate models and hypotheses [68]. This versatility makes SEM an invaluable tool for studying how factors such as fear-inducing elements and game design features contribute to player engagement and psychological responses. By applying SEM in this study, the dynamic interplay between external stimuli, internal processes, and resultant player behaviors can be systematically analyzed. According to Dash and Paul [73], SEM’s ability to model both direct and indirect effects simultaneously ensures a comprehensive understanding of the phenomena. It was added that PLS-SEM is deemed more sensitive than its counterpart, the covariance-based SEM, making it an indispensable tool for uncovering the dynamics of fear and engagement in the context of horror gaming.

4. Results

4.1. Results of Initial SEM

The initial SEM confirmed that all 12 hypotheses were statistically significant. This is because all p-values were below the 0.05. However, some of the factor loadings of the items reflecting the variables were deemed insignificant (less than 0.70). These are suggested to be removed to enhance the overall representation and measurement of the model and the overall model fit [68,73] (See Fig. 2).

Out of 26 indicators, 12 exhibited factor loadings below the 0.70 threshold and were removed from the PLS-SEM analysis. A refined set of 17 indicators remained, as shown in Table 2, after the removal of insignificant indicators. The retention of these indicators strengthens the measurement model, ensuring that only the most reliable items contribute to the assessment of each construct. Moreover, the results of the convergent validity analysis, demonstrating the reliability and internal consistency of the study’s constructs. Construct reliability was assessed using Cronbach’s alpha (CA) and Composite Reliability (CR), both of which exceeded the recommended threshold of 0.70 [73]. Their average variance extracted (AVE) were also above the recommended threshold of 0.50 [68]. These findings confirm that the constructs exhibit strong internal consistency, ensuring unbiased and reliable participant responses. This creates the final SEM (Fig. 3) that represents actual gameplay on horror video games.

In PLS-SEM, ensuring the distinctiveness of constructs is essential for

Table 2
Descriptive Analysis of Indicators.

Indicators	Mean	Standard deviation	Shapiro-Wilk	Initial FL	Final FL	CA	CR	AVE
FI1	4.049	0.853	1.860	0.761	0.818	0.782	0.794	0.659
FI2	3.823	0.810	0.784	0.564	—			
FI3	3.934	0.884	1.430	0.789	0.805			
GD1	3.840	1.020	0.481	0.684	—			
GD2	3.925	0.845	1.702	0.621	—			
GD3	3.931	0.848	0.686	0.712	0.791	0.870	0.821	0.697
GD4	3.996	0.857	1.195	0.720	0.876			
GD5	4.034	0.804	1.660	0.557	—			
PR1	4.230	0.841	−0.835	0.791	0.825			
PR2	3.860	0.879	1.132	0.787	0.749			
CH1	4.263	0.794	−0.415	0.736	0.788	0.792	0.766	0.621
CH2	4.060	0.820	−0.252	0.534	—			
CH3	4.008	0.849	0.811	0.764	0.810			
EV1	4.212	0.835	1.632	0.718	0.773			
EV2	4.044	0.772	1.408	0.641	—			
EV3	4.077	0.841	0.827	0.768	0.795	0.774	0.762	0.615
PC1	4.099	0.891	1.075	0.783	0.797			
PC2	3.967	0.784	0.722	0.528	—			
PC3	3.875	0.882	−1.197	0.767	0.817			
PT1	4.014	0.924	0.776	0.712	0.831			
PT2	4.137	0.750	1.138	0.708	0.807	0.764	0.789	0.651
PT3	3.839	0.905	1.156	0.617	—			
PT4	4.057	0.868	1.170	0.612	—			
AG1	4.091	0.901	0.924	0.797	0.812			
AG2	3.956	0.851	1.393	0.737	0.736			
AG3	3.940	0.922	1.206	0.805	0.790			

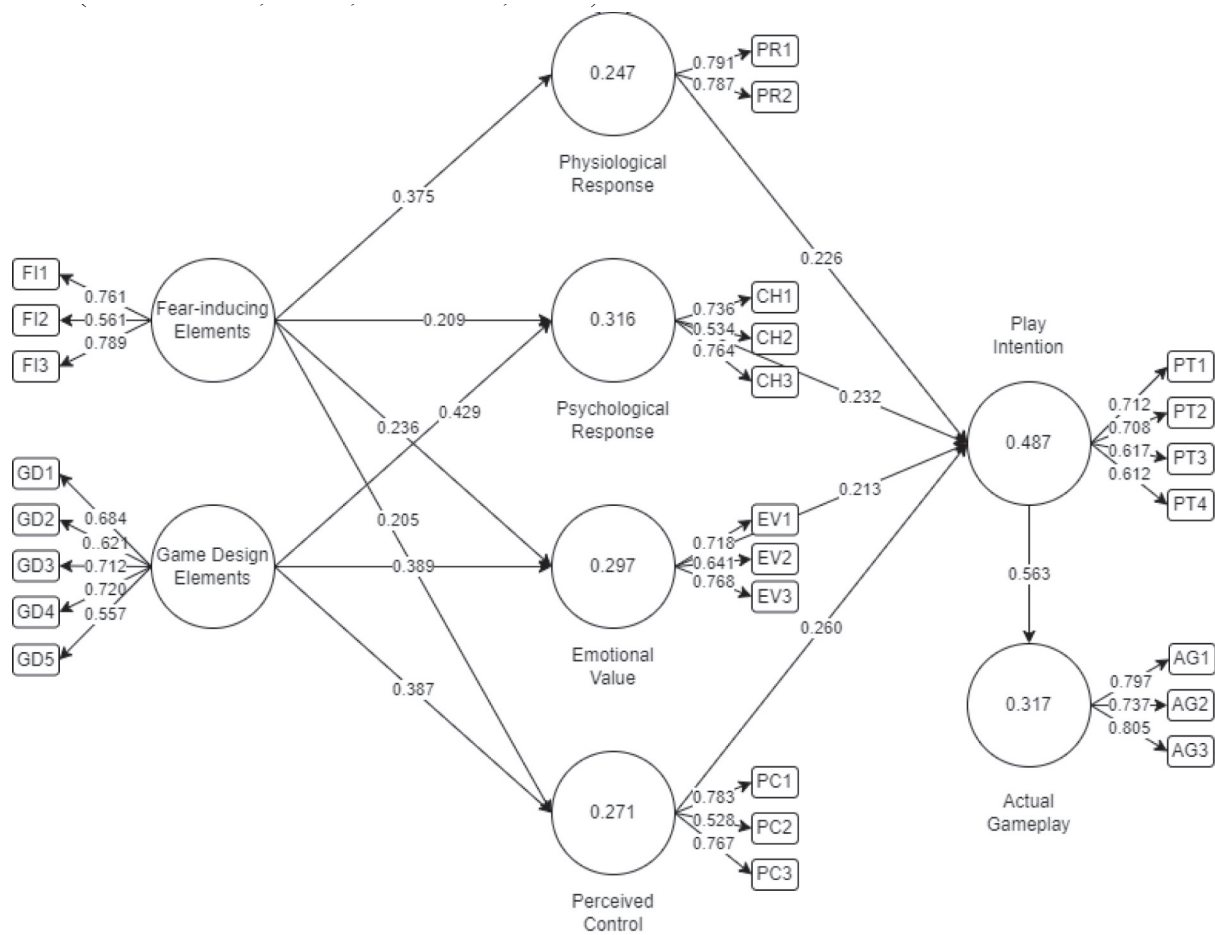


Fig. 2. Initial Structural Equation Modeling.

maintaining the validity of the measurement model. To assess this, the Fornell-Larcker criterion and the Heterotrait-Monotrait (HTMT) ratio were employed [74]. The Fornell-Larcker criterion determines whether each construct is more strongly related to its own indicators than to other constructs by comparing the square root of the AVE with the inter-construct correlations [75]. As shown in Table 3, the diagonal values are consistently higher than the corresponding off-diagonal values in their respective rows and columns. This confirms that each construct is accurately measured and does not exhibit excessive overlap with others, thereby establishing discriminant validity.

To further validate the constructs and prevent multicollinearity, the HTMT ratio was examined. A construct achieves discriminant validity when its HTMT value remains below the recommended threshold of 0.90 [76]. As presented, all HTMT values demonstrated that none of the constructs exhibit excessively high correlations. These findings affirm the structural integrity of the model, ensuring that each construct retains its unique contribution without redundancy. The results indicate that both reliability and discriminant validity are successfully established, supporting the robustness of the measurement model.

4.2. Model fit analysis

The model fit using three key measures further reinforces the acceptance of the final model: Standardized Root Mean Square Residual (SRMR), Chi-Square, and Normed Fit Index (NFI). The SRMR value of 0.072 falls within the recommended limit of less than 0.08 [77], indicating that the model accurately represents the observed data with minimal residual error. The Chi-Square value of 2.343 satisfies the required threshold of below 5.00 [78], reinforcing the model's

adequacy. Furthermore, the NFI score of 0.842 surpasses the minimum benchmark of 0.80 [78], demonstrating a strong fit compared to a baseline model. These results confirm that the proposed model aligns well with the data, ensuring its validity for further analysis.

Table 4 presents the summarized hypothesis testing results, proving the strength and significance of the relationships within the model. Among these, the path from Play Intention on Actual Gameplay ($\beta = 0.542$, $p < 0.001$) demonstrated the strongest effect, highlighting that a player's intention to engage with the game is a critical predictor of actual gameplay involvement. This strengthens the importance of fostering strong play intentions through immersive mechanics, compelling narratives, and well-balanced challenges to sustain player engagement [18,54].

Fear-Inducing Elements significantly impacted Physiological Response ($\beta = 0.417$, $p = 0.000$), Psychological Response ($\beta = 0.272$, $p = 0.000$), Emotional Value ($\beta = 0.251$, $p = 0.000$), and Perceived Control ($\beta = 0.229$, $p = 0.000$). These findings indicate that fear-inducing elements evoke strong physiological and psychological reactions while simultaneously enhancing emotional engagement and the player's sense of control over the gaming experience [16,37]. This suggests that integrating fear elements can effectively heighten immersion and player retention [50].

Game Design Elements also exhibit substantial effects, particularly on Perceived Control ($\beta = 0.364$, $p = 0.000$), Emotional Value ($\beta = 0.359$, $p = 0.000$), and Psychological Response ($\beta = 0.310$, $p = 0.000$). These results suggest that well-structured game design enhances a player's ability to manage in-game situations, elevates emotional connection, and influences psychological engagement [44,46].

Additionally, the paths from Physiological Response ($\beta = 0.085$, $p =$

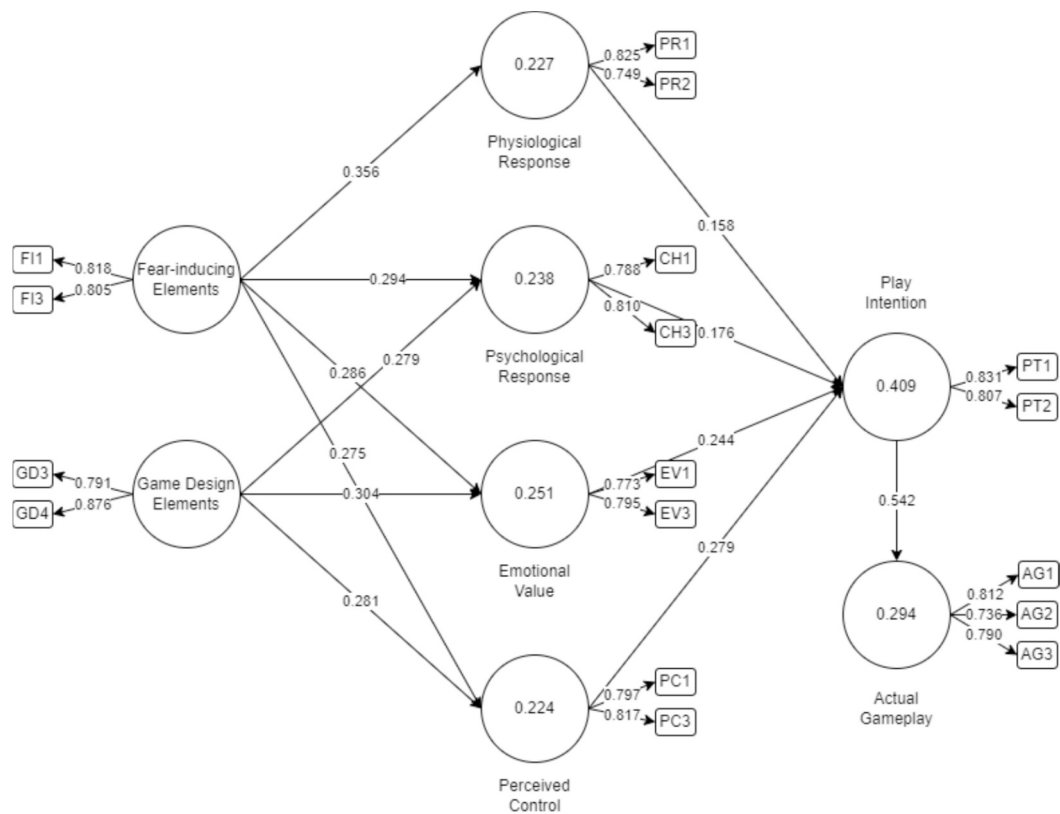


Fig. 3. Final Structural Equation Modeling.

Table 3
Discriminant Validity.

Fornell–Larcker Criterion								
	Actual Game Play	Psychological Response	Emotional Value	Fear-Inducing Elements	Game Design Elements	Perceived Control	Physiological Response	Play Intention
Actual Game Play	0.780							
Psychological Response	0.426	0.799						
Emotional Value	0.448	0.452	0.784					
Fear-Inducing Elements	0.456	0.419	0.421	0.812				
Game Design Elements	0.445	0.410	0.431	0.445	0.835			
Perceived Control	0.444	0.448	0.441	0.400	0.403	0.807		
Physiological Response	0.318	0.300	0.352	0.356	0.317	0.347	0.788	
Play Intention	0.542	0.458	0.501	0.459	0.435	0.519	0.393	0.819
Heterotrait-Monotrait Ratio								
	Actual Game Play	Psychological Response	Emotional Value	Fear-Inducing Elements	Game Design Elements	Perceived Control	Physiological Response	Play Intention
Actual Game Play	0.758							
Psychological Response	0.758							
Emotional Value	0.665	0.825						
Fear-Inducing Elements	0.785	0.715	0.788					
Game Design Elements	0.701	0.822	0.819	0.846				
Perceived Control	0.771	0.799	0.759	0.726	0.768			
Physiological Response	0.607	0.726	0.814	0.814	0.663	0.801		
Play Intention	0.802	0.772	0.848	0.726	0.796	0.765	0.775	

0.016), Psychological Response ($\beta = 0.176$, $p = 0.000$), Emotional Value ($\beta = 0.261$, $p = 0.000$), and Perceived Control ($\beta = 0.295$, $p = 0.000$) on Play Intention were all significant, reinforcing their collective role in shaping a player’s motivation to engage with the game. These results

affirm that a player’s experience—whether physiological, psychological, emotional, or through perceived control—directly influences their willingness to continue playing [22,50,53].

Table 4
Summarized Hypothesis Output.

Hypotheses	Relationship	Beta	P value	Decision
1	FI → PR	0.356	<0.001	supported
2	FI → CH	0.294	<0.001	supported
3	FI → EV	0.286	<0.001	supported
4	FI → PC	0.275	<0.001	supported
5	GD → CH	0.279	<0.001	supported
6	GD → EV	0.304	<0.001	supported
7	GD → PC	0.281	<0.001	supported
8	PR → PT	0.158	0.002	supported
9	CH → PT	0.176	<0.001	supported
10	EV → PT	0.224	<0.001	supported
11	PC → PT	0.279	<0.001	supported
12	PT → AG	0.542	<0.001	supported

5. Discussion

The findings revealed a strong positive relationship between Play Intention (PT) and Actual Gameplay (AG) ($\beta = 0.542$, $p = 0.000$), indicating that a player's intent to engage significantly influences their actual participation. This aligns with Flanagan [54], who found that players with strong play intentions are more likely to invest time and effort, engage in immersive gameplay, and persist through challenges, whereas those with low play intention tend to disengage more easily. Similarly, Christopher and Leuszler [18] observed that in horror games, players with higher play intention were more inclined to interact with objects, explore environments, and engage deeply with the narrative, reinforcing the idea that intention translates into active gameplay. Marak [55] further supports this by demonstrating that players with strong play intention exhibited greater problem-solving abilities and heightened immersion. However, while PT is the strongest predictor of AG, its antecedents vary in predictive strength, suggesting that different psychological and design factors influence a player's decision to engage, which may shape how intention translates into actual behavior across different game genres and contexts.

Building on the strong relationship between PT and AG, additional studies further underscored the significance of this connection. Hattingh et al. [79] found that multiple factors, including habit, price-value, flow, and effort expectancy, significantly influence ongoing gameplay intentions among amateur eSports gamers, reinforcing the idea that a player's intent to engage is shaped by both intrinsic motivation and external conditions. Their findings suggested that when players perceive gaming as habitual or valuable, their play intention strengthens, leading to sustained gameplay. This supports the notion that PT is not merely an abstract concept but is driven by cognitive and behavioral mechanisms that translate into actual gaming behavior. Meanwhile, Limantoro and Atma [80] emphasized the immersive nature of psychological horror games, which engage players by evoking deep emotional and cognitive responses. Their findings suggested that the psychological complexity of horror narratives intensifies player involvement, further validating that heightened play intention—particularly in emotionally charged and mentally engaging genres—leads to more active and sustained participation.

Among the determinants of PT, Perceived Control (PC) ($\beta = 0.295$, $p = 0.000$) emerged as the strongest, signifying that players who feel more in control of the game's mechanics and outcomes are more likely to intend continued play. This supports theories on autonomy and agency in gaming, where control enhances motivation. Verran [53] further validated this by showing that identification with avatars, such as in *Celeste*, enhances engagement through mastery and self-exploration, reinforcing the idea that control over in-game actions fosters deeper player investment. Similarly, Moschovitis and Denisova [28] found that biofeedback-controlled dynamic difficulty adjustment (DDA), as implemented in *Caroline*, increases perceived control by dynamically balancing challenge and mastery, leading to higher intrinsic motivation. Contradicting findings from other studies suggest that this effect may

not be universal across all game genres.

Kesuma and Princes [81] found that in mobile *gacha* games, PC was not a primary driver of continuance intention. Instead, structural video game characteristics (SVGC), such as presentation, reward, and control/manipulation features, significantly influenced perceived enjoyment, while factors like hedonic motivation, habit, and trust in developers had stronger effects on player retention. The discrepancy likely stems from the fundamental differences in game design—unlike skill-based horror and action games, where player agency is crucial for engagement, *gacha* games rely on randomized mechanics and extrinsic rewards, making control less relevant to sustained play. This distinction highlights that while PC is a key determinant in interactive, skill-based games, its impact diminishes in game genres where unpredictability and reward anticipation drive player motivation.

Similarly, Liao et al. [82] challenged the role of perceived control by emphasizing the impact of real-world frustration on game engagement. Their study found that autonomy and competence frustrations in daily life drive players toward escapism, which then fosters continued playing intention. Unlike our findings, which suggest that in-game control is a direct motivator, Liao et al. [82] indicate that players may be more influenced by external psychological stressors rather than their sense of mastery within a game. This suggests a distinction between escapism-driven engagement, where games serve as a coping mechanism, and control-driven engagement, where mastery and agency are central. The divergence in findings may be attributed to differences in player motivation—our study focuses on games where skill and agency play a crucial role, whereas Liao et al. [82] examine broader patterns of game usage driven by external frustrations.

Interestingly, Emotional Value (EV) ($\beta = 0.261$, $p = 0.000$) closely follows, showing that emotional immersion significantly impacts play intention. While both PC and EV strongly influenced PT, their effects diverge: PC relates to cognitive engagement through a sense of mastery, whereas EV is more affective, stemming from excitement, fear, or joy. EV in horror games extends beyond fear, encompassing deep emotional engagement through storytelling, character empathy, and immersive design [22,33,35]. Vivekanandhan et al. [52] found that horror games elicit heightened cognitive engagement compared to calmer genres, fueling excitement and sustaining attention, which, in turn, increases play intention. Additionally, Tomé et al. [22] demonstrated that players' emotional connections with characters in narrative-driven horror games create experiences that encourage game revisits, reinforcing the link between emotional value and sustained play.

Despite the strong influence of EV on play intention, some studies challenge the extent to which emotional immersion contributes to sustained gaming engagement. Greitemeyer [83] argued that video game consumption, particularly violent games, tends to increase aggression while decreasing prosocial behavior, suggesting that the emotional effects of gaming may not always lead to positive engagement. This contradicts the idea that emotional connections with characters and narratives necessarily enhance long-term play intention. Additionally, Goagoses et al. [84] highlighted that prolonged gaming may be more closely associated with maladaptive emotion regulation strategies, such as catastrophizing and rumination, rather than positive emotional engagement. Their findings suggest that some players may turn to gaming not for immersive storytelling or character attachment but as a means of coping with stress and negative emotions, which can lead to problematic gaming rather than sustained voluntary play.

Psychological Response (CH) ($\beta = 0.176$, $p = 0.000$) and Physiological Response (PR) ($\beta = 0.085$, $p = 0.016$) exhibited weaker but still significant effects on PT. The lower effect of PR suggests that while physical reactions to fear-inducing elements (FI) exist, they do not directly translate into sustained engagement as strongly as cognitive and emotional factors. This contrasts with conventional horror game theories that often emphasize physiological arousal as a primary driver of immersion. Instead, this study suggests that cognitive processing (CH) plays a larger role in reinforcing PT than bodily reactions (PR). Yan [17]

supports this by demonstrating that psychological responses—such as fear, anxiety, excitement, and immersion—significantly influence play intention, with strong emotional reactions motivating players to continue for escapism and adrenaline.

Similarly, Zhang [50] found that thrill-seekers find intense emotions rewarding, reinforcing the idea that psychological engagement sustains play, whereas others may withdraw if the fear becomes overwhelming. This aligns with Hawco [49] who highlighted how elements like haptic mechanics, play conditioning, and coding heighten fear and tension, create an immersive experience that can either be addictive or deter engagement, depending on individual tolerance for fear. Additionally, Luong and Holz [51] explored how transient psychological states, such as fear and frustration, trigger distinct physiological responses that influence engagement, further demonstrating that while PR plays a role, it is the cognitive and emotional responses that primarily drive sustained play intention. These findings suggest that while physiological reactions contribute to immersion, it is the player's cognitive processing of fear and excitement that has a stronger impact on their decision to continue playing.

Analyzing the antecedents of PC, EV, and CH, Game Design Elements (GD) appeared more influential than Fear-Inducing Elements (FI). GD significantly impacts PC ($\beta = 0.364$, $p = 0.000$) and EV ($\beta = 0.359$, $p = 0.000$), surpassing the effects of FI on these factors. This suggests that structured game mechanics, balanced challenges, and interactive design features shape both perceived control and emotional engagement more effectively than fear stimuli alone. Zhang et al. [33] support this by highlighting the crucial role of game mechanics in shaping psychological responses to horror games. Smith [21] further proved that elements such as atmospheric design, sound, and ambiance elicit strong emotional reactions, particularly tension and stress, reinforcing the idea that immersive game design enhances engagement. Similarly, Qi and Jeanhuan [44] emphasized the impact of character design in side-scrolling horror puzzle games, showing that effective integration of gameplay scenarios enhances emotional experiences and psychological responses. While fear-based stimuli disrupt players' sense of control [13,34,35,42], well-crafted game mechanics provide structured engagement that fosters immersion.

Furthermore, specific game design elements such as lighting, sound, and environmental design significantly influence psychological responses in horror games, which developers could enhance. Cao et al. [45] and Griffith and Sharpe [14] found that players in dimly lit, eerie environments reported lower perceived control compared to those in brighter, open settings, demonstrating that environmental design affects player agency and immersion. Fouché [46] extended this discussion by revealing that minimalist design, obfuscation, and clumsy mechanics heightened unease and immersion. These findings suggested that while complex mechanics enhance engagement, intentionally limited control can amplify tension, reinforcing that a player's sense of agency remains central to play intention. Additionally, Lemmens et al. [43] and Cao et al. [45] found that high-quality sound effects and music further reduced perceived control, amplifying emotional tension. Collectively, these studies suggest that while FI elements contribute to fear and anxiety, GD elements are more instrumental in sustaining engagement by balancing control, challenge, and immersion.

Surprisingly, FI exhibit the weakest effect on PC ($\beta = 0.229$, $p = 0.000$), highlighting a contrast between fear's role in immersion versus control. While fear-inducing elements such as eerie sound effects, jump scares, and unsettling visuals are designed to evoke heightened emotional and physiological responses [33–35], they do not necessarily empower players with a sense of agency. This aligns with findings by Ferreira et al. [42], which suggest that while players can feel a degree of control when navigating fear-based scenarios, unpredictable threats and disorienting mechanics often diminish this perception. Similarly, Lemmens et al. [43] demonstrated that while players' perceived control over the game environment can mitigate fear, the sudden loss of control—such as being ambushed or trapped—intensifies anxiety and

tension.

Fear's impact on CH ($\beta = 0.272$, $p = 0.000$) further reinforces that while fear-inducing elements provoke cognitive responses, their influence on psychological engagement remains secondary to GD elements. Studies by Tu and Mozgovoy [41] and Tomé et al. [22] confirmed that unpredictable encounters and terrifying entities disrupt the player's sense of stability, fostering cognitive tension rather than sustained engagement. However, FI exerts the strongest effect on PR in this study ($\beta = 0.417$, $p = 0.000$), emphasizing its role in triggering physiological arousal. This supports research by Németh and Szűcs [36], who found that horror and violent games elicit measurable changes in heart rate and blood pressure compared to non-fear-based games. Furthermore, Kovács [37] explored the “horror paradox,” demonstrating that controlled fear responses—where players experience short bursts of terror followed by relief—can generate a pleasurable dopamine release, reinforcing fear's ability to evoke strong physiological reactions.

However, since PR exerts the weakest influence on PT, the findings suggest that while fear-based stimuli effectively create immediate physical arousal, they do not translate into long-term play motivation as strongly as cognitive and emotional factors. This aligns with Andersen et al. [11], who observed an inverted U-shaped relationship between fear and enjoyment, with moderate fear yielding the most optimal engagement. Waili [16] similarly noted that while fear-inducing elements heighten alertness and tension, excessive fear can overwhelm players, reducing overall enjoyment and sustained play interest. Consequently, while fear is instrumental in shaping momentary immersion, its capacity to drive continued engagement remains dependent on factors such as game mechanics and narrative depth [22,41].

Overall, the study highlights a contrast between the effects of GD and FI elements. GD is more influential in shaping sustained player engagement through control, psychological involvement, and emotional value, whereas FI primarily impacts physiological reactions and, to a lesser extent, emotional and psychological responses. This suggests that while horror elements can enhance immersion, they alone do not drive play intention as strongly as well-designed game mechanics. Therefore, a balanced integration of engaging gameplay mechanics with immersive fear-based elements may be the most effective strategy for sustaining player engagement.

5.1. Theoretical Implication

The self-structured S-O-R framework serves as a powerful theoretical tool in this study, allowing for a comprehensive examination of how horror games trigger adrenaline and sustain fear appraisals from fear-inducing elements. Unlike traditional models such as the Technology Acceptance Model (TAM) or the Uses and Gratifications Theory (UGT), which primarily focus on motivation and perceived usefulness [85], this study's SOR model captures the dynamic psychological and physiological mechanisms at play in horror gaming. This is particularly significant because horror game engagement is not solely driven by rational decision-making or functional utility, but is deeply tied to visceral emotional responses and subconscious cognitive processing. By integrating FI and GD elements as external stimuli, this study advances the application of SOR beyond consumer behavior, demonstrating its capacity to explain the intricate relationship between game design, emotional arousal, and player behavior [30,32].

A major contribution of this study is its extension of the SOR framework to fear-inducing elements in gaming, an area overlooked in previous applications of the model and elements among related studies. While past studies have utilized SOR in digital media and entertainment, such as investigating user engagement in Extended Reality (XR) gaming [32] or analyzing how emotional stimuli influence purchasing behavior [31], this research is among the first to empirically validate how horror games manipulate fear responses to maintain engagement and reinforce addictive play behaviors. Unlike linear models such as TAM, which emphasize voluntary adoption based on perceived ease of use and

usefulness, SOR allows for a multi-layered analysis of involuntary emotional reactions that drive player immersion, repeated exposure, and sustained play intentions. This highlights the framework's superiority in capturing the interplay between fear, excitement, and behavioral persistence in horror gaming.

Additionally, by employing PLS-SEM, this study enhances the predictive power of the SOR framework, offering a quantitative validation of the relationships between fear-inducing stimuli, emotional responses, and actual gameplay behavior. The framework's modular structure, which organizes stimuli, organismic responses, and behavioral outcomes into distinct yet interconnected constructs, allows for a precise examination of psychological mechanisms that simpler models fail to capture. This structured approach not only strengthens the theoretical foundations of horror game research but also offers valuable insights for game developers, psychologists, and scholars investigating the intersection of gaming, fear, and behavioral addiction. Through these contributions, this study establishes the SOR framework as an essential model for understanding the complex emotional and psychological drivers of horror game engagement.

5.2. Practical and Managerial implications

5.2.1. For game designers and developers

The findings of this study offer crucial insights into how horror games trigger adrenaline and contribute to fear appraisals, helping developers design more immersive, psychologically balanced, and safe experiences. By understanding the interplay between fear-inducing stimuli, physiological and psychological responses, and player engagement, game designers can refine horror mechanics to enhance both thrill and enjoyment while mitigating potential negative effects of excessive fear or stress. This study highlights the importance of incorporating adaptive difficulty settings that adjust fear intensity based on player responses, fear pacing techniques that provide moments of relief between intense sequences, and strategic moments of perceived control, such as allowing players to temporarily disable threats or provide safe zones, to prevent overwhelming anxiety.

Additionally, developers can implement gradual exposure mechanics, where fear elements escalate progressively, ensuring players are not immediately overwhelmed. Another approach is customizable horror experiences, enabling players to adjust specific fear triggers, such as jump scares or enemy aggression, to suit their comfort levels. By ensuring that games remain thrilling yet psychologically safe, promoting overall well-being among players could be established. For example, developers can integrate biofeedback-based difficulty scaling, where heart rate monitors or player reaction data dynamically adjust game intensity, preventing excessive stress while maintaining engagement. They can also design alternative escape mechanisms, like hiding-based survival rather than constant combat, reducing prolonged exposure to high-adrenaline fear states. By designing horror games that strike a balance between fear and enjoyment, developers can create experiences that maximize immersion without compromising player mental health.

5.2.2. For horror game players

This study provides valuable insights for horror game players by shedding light on how adrenaline rushes and fear-inducing elements influence their gaming experiences. Understanding these psychological and physiological mechanisms can empower players to make informed decisions about their gaming habits by setting personal limits on play duration, recognizing signs of excessive fear or stress, and using relaxation techniques such as deep breathing or grounding exercises during intense moments. This approach ensures that players engage with horror games in ways that enhance enjoyment without causing excessive anxiety.

The study also emphasizes the role of perceived control in shaping fear experiences, suggesting that players can optimize their experience by adjusting difficulty settings to match their comfort level, playing in

well-lit environments to reduce immersion intensity, or using audio modifications to lessen fear-inducing sound effects. Engaging with horror games strategically—such as by pacing gameplay sessions or gradually exposing themselves to scarier elements—can also help build emotional resilience while maintaining a positive gaming experience.

Furthermore, by recognizing how fear-based engagement affects emotional and cognitive responses, players can harness the psychological benefits of horror games. These benefits include improved problem-solving skills through puzzle-solving mechanics in horror games, enhanced resilience under pressure by overcoming high-stress scenarios, and heightened situational awareness by learning to anticipate threats and react accordingly. These cognitive and emotional developments, which promotes learning and mental adaptability through gaming could be aligned.

5.2.3. For gaming industry marketers and publishers

Marketing strategies for horror games can be enhanced and capitalized mindfully by leveraging the study's findings on fear-based engagement and adrenaline-driven experiences. The study suggests that fear-inducing elements plays a significant role in player retention, making it a crucial factor in-game marketing. Marketers can develop promotional strategies that emphasize the thrill-seeking aspects of horror games, targeting audiences who actively seek adrenaline-fueled experiences. Interactive horror game demos, psychological thrill challenges, and immersive VR horror experiences can serve as effective promotional tools. However, responsible marketing should also consider potential psychological impacts, ensuring that games are positioned as thrilling yet manageable experiences rather than sources of excessive fear and distress. It is posited that the innovative actions in technology could help establish the gaming industry while also bridging the gap among the digital divide among consumers. By aligning marketing strategies with evidence-based psychological insights, the gaming industry can contribute to SDG 9 by fostering innovation and sustainable growth in immersive gaming technologies.

5.3. Limitations and future research

Despite the valuable insights provided by this study, several limitations must be acknowledged, which future research could address to further enhance the understanding of adrenaline rushes, fear-inducing elements, and their impact on gaming psychology. Future studies could incorporate physiological measurements, such as heart rate variability or galvanic skin response, to obtain more objective data on fear responses and adrenaline activation – actual experimentation and physical ergonomics assessment. Moreover, it is suggested to consider negative implications, ambivalent reactions, and or countermeasures for more holistic assessment of both positive and negative insights since this benchmark study only considered positive reflections. In accordance, future research may opt to individually measure some of the items like GD4, PC4, and PR2 since these could be measured individually and may vary among respondents. Additionally, this study focuses primarily on digital horror games, potentially overlooking the psychological effects of other horror media, such as VR horror experiences or mixed-reality games, which may elicit different levels of immersion and fear responses. Exploring these alternative gaming formats could provide a more comprehensive understanding of how fear-based engagement varies across different platforms. While this study examined fear-inducing elements and its effects on player engagement, it does not extensively analyze potential long-term psychological effects, such as desensitization to fear or anxiety regulation.

This study also utilized a structured questionnaire, adapted from related studies. Despite its establishment on related literatures, reliability, and validity, there are still areas of potential inherent bias among self-reported data and potential inaccuracies. Future research could adopt a longitudinal approach to investigate the prolonged impact of fear-based gaming on cognitive and emotional well-being, ensuring

that gaming remains both engaging and psychologically safe. In addition, future studies may consider the utilization of mixed method approach of qualitative and quantitative analysis from interviews, group discussion, and assessments from survey to holistically measure the intended game design elements, psychological, and physiological player response assessment among horror games. It could also be deduced that the comparison of player types, both between horror and non-horror genre players may be accounted by future studies. Since this was established as a benchmark study, the current analyses involved horror genre-aligned players and may have caused the parallel positive output. The bias that the overall measurement has due to positive and actual horror game players, and the consideration only of established horror games that has been meticulously designed and established should be compared and contrast among their counterparts by future studies.

6. Conclusion

This study explored the impact of the stimulus, FI and GD elements on player experience, using data from 1015 respondents and analyzing relationships through PLS-SEM with SOR-based analysis. It was revealed that FI significantly impact both CH and EV, demonstrating that fear-driven stimuli evoke strong cognitive and emotional reactions from players. In contrast, GD have a more pronounced effect on PC and PT, suggesting that structured design elements contribute more to a player’s sense of agency and willingness to engage with the game. This distinction highlights how FI and GD shaped different aspects of player experience—FI enhances immersion by triggering emotional arousal and psychological engagement, while GD fosters sustained participation by reinforcing structured mechanics and player control.

Interestingly, CH is strongly associated with EV, indicating that the cognitive processing of fear enhances the emotional significance of the game. However, PR, while also linked to EV, exhibited a weaker correlation. This suggests that while both psychological and physiological reactions contribute to emotional experiences, the cognitive interpretation of fear plays a more dominant role in shaping how players perceive the emotional impact of the game. This finding contradicts traditional theories emphasizing physiological arousal as the primary driver of emotional engagement.

The relationship between EV and PC presents an interesting contrast. While high EV is generally associated with positive engagement, excessive emotional intensity (especially from FI) can diminish PC,

leading to reduced player agency and control. On the other hand, GD enhance PC, ensuring that even in fear-inducing scenarios, players feel capable of navigating challenges. This suggests a balance is necessary: while fear elements heighten emotional value, well-designed mechanics must maintain a sense of control to sustain engagement.

The results indicate that PT is a strong predictor of AG, but discrepancies arise when FI are highly intense. In such cases, even if PT is high, AG may be lower due to overwhelming fear or distress, which hinders sustained gameplay. Conversely, when GD are prominent, PT translates more directly into AG, reinforcing the importance of balancing fear elements with engaging mechanics to prevent player dropout.

Overall, these findings can guide game designers, players, and marketers. Developers can refine game mechanics to balance fear and player agency, enhancing immersion while minimizing anxiety. Players can optimize their experience by managing difficulty and coping strategies, while marketers can responsibly promote horror games as thrilling yet psychologically safe. Additionally, exploring other immersive formats like VR horror and adopting a longitudinal approach could provide deeper insights into the long-term psychological effects of fear-based gaming, further advancing both game design and player well-being.

CRediT authorship contribution statement

Klarisse Nicole N. Savino: Data curation, Conceptualization, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Writing – review & editing, Writing – original draft. **Ronila Angela B. Mateo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ardvin Kester S. Ong:** Supervision, Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Measure Items and Indicators.

Variables		Indicator	Constructs	References
Stimulus (S)	Fear-Inducing Elements	FI1	The use of eerie sounds and background music in horror games intensifies my feelings of fear and suspense.	Steinhaeusser et al. [34], Zhang et al. [33], Villarreal [35]
		FI2	Dark and grotesque visual imagery in horror games effectively triggers feelings of fear.	
		FI3	Jump scares in horror games startle me and enhance my overall gaming experience by making it more thrilling.	
	Game Design Elements	GD1	The immersive graphics in horror games significantly enhance the fear-inducing experience by creating a more realistic and engaging environment.	Steinhaeusser et al. [19], de Lima et al. [13], Zhang et al. [33], Waili [16], Villarreal [35], Ferreira et al. [42]
		GD2	The unpredictability of pacing in horror games keeps me on edge, heightening my overall sense of fear and anticipation.	
		GD3	The storyline in horror games engage me emotionally, making the gameplay more immersive and frightening.	
		GD4	The character development in horror games engage me emotionally, making the gameplay more immersive and frightening.	
		GD5	Horror games evoke strong feelings of fear that keep me mentally engaged, even when there is no immediate danger in the game.	

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	Variables	Indicator	Constructs	References
Organism (O)	Physiological Response	PR1	While playing horror games, I experience an increase in heart rate or palpitations due to fear or suspense.	Hanshans et al. [20], Villarreal [35]
		PR2	I feel a surge of adrenaline in my body when encountering fear-inducing elements, such as jump scares or tense situations, in horror games.	
	Psychological Response	CH1	Playing horror games heightens my sense of alertness, making me more aware of my surroundings and the game's events.	Perez-Reyes [40], Zhang et al. [33], Yi and Chung [39]
		CH2	I seek out horror games specifically to experience the thrill and excitement that comes from fear-inducing moments.	
	Emotional Value	CH3	Horror games heighten my five senses during gameplays.	Khunnasut [23], Villarreal [35], Zhang et al. [33]
		EV1	Overcoming fear-inducing challenges in horror games gives me a sense of excitement and exhilaration.	
		EV2	I feel a sense of satisfaction and achievement after facing fear-inducing elements in a horror game.	
	Perceived Control	EV3	The emotional thrill of surviving fear-inducing moments in horror games is one of the main reasons I enjoy playing them.	Andersen et al. [11], Lemmens et al. [43]
		PC1	I believe I can navigate through fear-inducing situations in horror games with ease.	
		PC2	I feel confident in my ability to manage fear and overcome challenges while playing horror games.	
Response (R)	Play Intention	PC3	I believe that my actions and decisions during gameplay significantly influence my ability to survive and succeed in horror games.	Tomé et al. [22], Smith [21], Villarreal [35]
		PT1	The adrenaline rush I experience while playing horror games makes me want to play them again.	
		PT2	I often look forward to playing more horror games after finishing one, driven by the fear and thrill they provide.	
		PT3	Playing horror games cause me to experience physical reactions such as sweating or feeling tense.	
	Actual Gameplay	PT4	The strong emotional feeling I experience creates an intensified the game feels.	de Lima et al. [13], Steinhäusser et al. [34], Zhang et al. [33], Villarreal [35]
		AG1	My instinctive reactions to fear-inducing elements (e.g., jump scares or sudden threats) drive me to stay alert and make quick decisions during gameplay.	
		AG2	The challenges and intense moments in the game compel me to strategize and adapt my actions to overcome obstacles and progress further.	
		AG3	The adrenaline rush I experience during gameplay motivates me to take on riskier actions or push through fear-inducing moments to achieve in-game goals.	

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.entcom.2025.101001>.

Data availability

Data will be made available on request.

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